Deena A.C

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Summary

Embedded Software Engineer with a B.Tech in CSE (IoT) and hands-on expertise in embedded C/C++, ESP32/microcontroller architectures, and real-time firmware development. Proven track record in developing low-power, deterministic systems, sensor interfacing (I2C, SPI, UART), and edge-AI integration, including an Indian patent for an edge-vision automation system that boosted energy efficiency by 27%. Published researcher in IEEE and Springer and IEEE RAS Hackathon award winner, skilled in bridging bare-metal hardware control, RTOS constraints, and intelligent edge processing for scalable industrial IoT and embedded applications.

Patents & Publications

Intellectual Property India Patent: ***Vision Track Systems***
Developed and patented published application Vision Track Systems (Filed Apr 2026, Published Jun 2026). Patent process completed 2 major steps (Filing & Publication), currently awaiting Request for Examination.
IEEE GCWCN 2025: *Adaptive Non-Intrusive Load Monitoring with Human-in-the-Loop Machine Learning for Appliance Disaggregation.*
Springer ICICCD 2025: *Intelligent Building Automation System using YOLOv8 and ESP32 for Energy-Efficient Occupancy Detection.*
IEEE ICCSC 2024: *Experimental Investigation for Hybrid Energy Harvesting System for Remote Monitoring Applications.*

Projects

Adaptive NILM with Human-in-the-Loop Machine Learning *Jul 2025 – May 2026*

- Developed a Non-Intrusive Load Monitoring (NILM) architecture leveraging hardware sensors to continuously disaggregate load profiles and extract individual device power signatures.
- Achieved 23% higher energy savings versus conventional NILM systems by reducing misclassification rates through real-time user verification and incremental model retraining.

Tech Stack: ESP32, PZEM-004T, Python, Machine Learning (Active Learning/Classification, Rule based Algorithm), Embedded C

Intelligent Building Automation using YOLOv8 and ESP32 *Jul 2025 – Dec 2025*

- Developed a vision-based building automation system using YOLOv8 on ESP32 microcontrollers to dynamically control HVAC and lighting based on real-time occupancy detection.
- Achieved 27% energy savings and significantly reduced false-positive triggers compared to traditional PIR sensor-based systems, which relied on motion detection rather than actual occupancy presence.

Tech Stack: YOLOv8, OpenCV, Python, ESP32, Wi-Fi/HTTP, Embedded C

Experience

Electric Vehicle Intern – Kodacy *June 2026 - July 2026 (Remote)*
Gained foundational expertise in EV architectures, classifications, and sustainable mobility solutions. Analyzed core subsystems including electric motors, power electronics (converters/inverters), Battery Management Systems (BMS), and transmission controllers. Evaluated regenerative braking, charging protocols, and e-differentials through practical simulations to optimize energy flow and performance

Education

B.Tech – Computer Science and Engineering (IoT) *2022 – 2026*
Kalasalingam Academy of Research and Education, Virudhunagar, Tamilnadu
CGPA: **8.53**
Higher Secondary School *2021 – 2022*
Fusco's Matriculation Higher Secondary School, Madurai, Tamilnadu
Score: **81%**

Skills

Languages & Tools: C, C++, Embedded C, Arduino IDE, KiCad (PCB - Beginner), SQLite (Beginner)
Microcontrollers: ESP32, ESP8266, Arduino
Protocols: I2C, UART, SPI, MQTT
IoT Cloud: Blynk, ThingSpeak, Arduino IoT Cloud
Edge AI: YOLOv8 Deployment, Sensor Fusion, Real-Time Data Acquisition
Core Expertise: System Architecture, Firmware Development, Circuit Design, Energy Optimization, NILM

Achievements

IEEE Conference Presenter • Springer Publication • IEEE RAS Hackathon – 2nd Prize for technical innovation • Ignite India (Wadhwani Foundation)

Certifications

Java Full Stack – Wipro TalentNext (Completed)

Embedded Systems with Arduino – GeeksforGeeks (Completed)

Introduction to C — C++ Intermediate – Sololearn (Completed)